

# Appendix A — Retrieval Benchmark and Vector-Store Selection

## Track B: RAG Engineering, Interoperability Services, and the Coordination-Layer Experiment

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### Scope of This Appendix

Phase 1 requires choosing a vector store for the retrieval service that Tier 2 will call during context assembly, before it reasons about ambiguous scenarios S4–S6. This appendix compares three open-source candidates — Chroma, FAISS, and Qdrant — against the project’s own measured benchmark (Section 2) and against figures reported in outside benchmarking work (Section 3), then applies both to the project’s actual requirements (Section 1 and Section 5).

*Revision note: this version supersedes the original Appendix A recommendation. The underlying benchmark data, external sources, and P99 analysis are unchanged; only the weighting applied to those results in Section 5 has been revised, and the final recommendation now selects Qdrant rather than Chroma. See Section 5 for the reasoning.*

### 1 Evaluation Criteria

Four criteria were used, in the order the project cares about them:

- C1. Average latency as a function of corpus size** — since the patient-profile knowledge base is small today but the retrieval service must remain usable as more scenarios and patients are added.
- C2. Install effort** — how much setup, configuration, and operational overhead (embedded library vs. standalone server) each store demands from a two-person, nine-week project.
- C3. Persistence to disk** — whether a store keeps its index across restarts natively, or needs an external persistence layer bolted on.
- C4. Tail latency, specifically P99** — discussed on its own in Section 4, because it is the metric that matters once retrieval sits on the critical path of a reasoning tier, even a tier that is not hard-real-time.

### 2 Our Own Benchmark

The measurements below come directly from the project’s own benchmark run (`results.csv`), reproduced in Table 1. Corpora of 6, 100, and 1000 synthetic patient-profile documents were indexed with each store, and three queries were issued against each corpus size.

Two patterns are visible in our own numbers. First, FAISS is the fastest store to build and to query at every corpus size tested, by roughly one to two orders of magnitude over Chroma and Qdrant. Second, Qdrant’s average latency grows the fastest of the three between 100 and 1000 documents (1.148 ms to 3.485 ms), while Chroma grows more gently (1.978 ms to 2.439 ms) and FAISS barely moves. At the scale this project actually needs — a knowledge base of a few dozen to a few hundred patient-profile snippets — all three stores answer in low single-digit milliseconds, so raw speed alone should not decide the choice.

A caveat that matters for Section 4: each corpus size in our own run was queried only three times, which is enough to see the mean but far too small a sample to estimate a 99th-percentile

Table 1: Own benchmark: index time and average query latency by corpus size.

| Store  | Documents | Index Time (s) | Avg. Latency (ms) | Queries |
|--------|-----------|----------------|-------------------|---------|
| Chroma | 6         | 0.1209         | 1.920             | 3       |
| Chroma | 100       | 0.1541         | 1.978             | 3       |
| Chroma | 1000      | 0.4140         | 2.439             | 3       |
| FAISS  | 6         | 0.00008        | 0.069             | 3       |
| FAISS  | 100       | 0.00013        | 0.066             | 3       |
| FAISS  | 1000      | 0.00058        | 0.203             | 3       |
| Qdrant | 6         | 0.0123         | 0.828             | 3       |
| Qdrant | 100       | 0.0365         | 1.148             | 3       |
| Qdrant | 1000      | 0.2636         | 3.485             | 3       |

Table 2: Own benchmark: qualitative comparison (from Appendix A source material).

| Store  | Install effort | Persists to disk                                          | Metadata filter     | Notes                                                                                                                   |
|--------|----------------|-----------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------|
| Chroma | Low–moderate   | Yes, with a persistent client and a persistence directory | Yes                 | Easy to prototype with locally; supports metadata-based patient lookup.                                                 |
| FAISS  | Low            | Not by default as a full vector database                  | No native filtering | Fastest raw search path, but needs a separate metadata store for <code>patient_id</code> filtering and for persistence. |
| Qdrant | Moderate       | Yes, when run with storage enabled                        | Yes                 | Combines vector search and payload filtering in one system.                                                             |

latency. Our own data therefore answers “how fast on average” but not “how bad in the worst case,” which is why the literature review below is used specifically to fill that gap.

### 3 Cross-Check Against External Benchmarks

Because our own run used a small, synthetic corpus and few queries, we cross-checked the direction of the results against independently published benchmarks that test the same three stores (plus related engines) at larger scale. These are industry or community benchmarks rather than peer-reviewed papers, so they are used only to confirm the *ranking* our own numbers already suggest, not as a substitute for the project’s own measurements.

The external sources agree with our own benchmark on the ordering at small scale (FAISS fastest, Chroma and Qdrant close behind) and add one useful warning our own run could not surface: Chroma’s advantage over Qdrant on raw latency seems to narrow or reverse as the corpus grows past roughly one million vectors [10]. That scale is far beyond anything this project’s patient-profile corpus will reach, so it is noted here as context. As Section 5 discusses, this appendix’s revised recommendation rests less on this scale argument and more on the concurrency behaviour documented in Section 4, which is directly relevant even at this project’s own small scale.

Table 3: Selected external benchmark figures, larger corpora (approximate, CPU-based unless noted).

| Source                                     | Scale                                      | Reported latency pattern                                                                                                                          | Relevant to this project                                                                                                                                 |
|--------------------------------------------|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| GigaGPU comparison [8]                     | 1M vectors                                 | FAISS on GPU reaches sub-millisecond single-query latency, several times faster than CPU-based Qdrant and Chroma at the same scale.               | Confirms FAISS’s raw-speed lead persists at scale, but the gap is largest only once GPU acceleration is used, which this project does not need.          |
| AIMultiple open-source vector DB study [9] | up to 2.25M vectors, MedRAG/TechQA corpora | Qdrant sustains a competitive multi-process throughput and a bounded p99 budget, while Chroma was excluded from the largest multi-process runs.   | The corpus used here (medical RAG) is close in spirit to our patient-profile use case.                                                                   |
| Qdrant vs. Chroma benchmark [10]           | 1M–10M vectors                             | Chroma’s own published benchmarks show p50 latency exceeding 400 ms at 1M vectors, while Qdrant stays under 10 ms at that scale.                  | Suggests Chroma’s latency advantage seen in our small-scale run would likely not hold if the knowledge base grew far beyond the current project’s scope. |
| HuggingFace FAISS/Chroma benchmark [11]    | Flickr8k, 6k items                         | FAISS shows the lowest average latency per query of the group tested, consistent with our own ranking.                                            | Independent confirmation of FAISS’s relative position using a different embedding model and dataset.                                                     |
| Qdrant official benchmarks [12]            | ann-benchmarks datasets                    | Qdrant reports the highest requests-per-second and lowest latency among the engines it compares itself against, across most precision thresholds. | A vendor-produced source; treated as directional evidence only, given the obvious incentive to favour its own tool.                                      |

## 4 Why P99, Specifically

The brief asked, specifically, why P99 rather than the average is the right latency metric to reason about here, so this section treats that question on its own.

### 4.1 The general argument

An average latency is computed across all requests and is easily dominated by the common case; a 99th-percentile (P99) latency is the response time below which 99% of requests fall, so it isolates the worst one-in-a-hundred request instead of hiding it inside a mean [5, 6]. Dean and Barroso’s widely cited systems paper *The Tail at Scale* makes the underlying point precisely: latency variability that looks negligible in a single component becomes dominant once many

components, or many repeated calls, are chained together, because the probability that *at least one* call in the chain lands in the slow tail grows quickly with the number of calls [1]. A system that only tracks its average latency can look healthy while a meaningful fraction of its users or requests are experiencing multi-second delays that the average never reveals [4].

## 4.2 Why it applies to this project even though Tier 2 is not hard-real-time

Tier 1 in this project’s architecture is the component with a genuine hard deadline, and it never calls the retrieval service — it never calls Tier 2 directly at all, by design (Project Architecture & Workflow Guide, §3–4). The retrieval service is called from Tier 2’s context-assembly step instead. That step does not need to finish in a fraction of a second, but it does sit on the path the reasoning model waits on before it can draft a severity/urgency judgement for scenarios S4–S6. Recent guidance on retrieval-augmented pipelines makes the same point in that specific setting: RAG systems depend on retrieving context before generation starts, so any variance in the retrieval step shows up directly in the pipeline’s own tail latency, even when the typical request is fast [7]. In other words, a retrieval call with an unremarkable *average* latency but an occasional multi-hundred-millisecond *tail* is exactly the failure mode that would make Tier 2’s output arrive late for the S4–S6 ablation runs, without ever showing up as a problem in the average-latency column of Table 1.

## 4.3 What this means for the choice of store

Two of the three candidates already show the shape of this risk in the data reviewed here. Qdrant’s own average latency in our benchmark grows faster with corpus size than Chroma’s or FAISS’s (Table 1), and external benchmarks show Chroma’s latency curve steepening sharply once a corpus grows into the millions of vectors [10]. Neither observation is visible from a single mean value at one corpus size, which is precisely why a P99 figure, not just an average, is treated as decisive in Section 5. Because our own benchmark used only three queries per corpus size, it cannot itself produce a meaningful P99; this is flagged as a concrete follow-up in Section 6 rather than glossed over.

Figure 1 makes the point concrete using the best-documented P99 figures we could find in the outside literature for two of our three candidates. No single published source benchmarks Chroma, FAISS, and Qdrant together on P99 at matched scale and load, so the figure combines separately-cited measurements rather than a single apples-to-apples run; each bar is labelled with its own scale and load condition, and this heterogeneity is deliberate rather than hidden. Two things stand out. First, Qdrant’s own P99 at a much larger scale (38.71 ms at 50 million vectors, 99% recall) [13] is still an order of magnitude below one second, and only about 16 times our own small-scale mean at 1000 documents (3.485 ms) — consistent with Qdrant’s latency degrading gracefully as scale and load grow, matching the qualitative claim in Section 3. Second, Chroma’s P99 under concurrent load in an independent benchmark reaches roughly 13 seconds at 512 simultaneous requests, because its HNSW parameters are fixed at collection-creation time and it does not scale gracefully under concurrency [9]. That figure is around 5,000 times larger than our own sequential-query mean for Chroma at the same corpus scale (2.439 ms) — the single clearest illustration in the sources we reviewed of why an average, on its own, would hide a real operational risk. This is also the specific data point the revised recommendation in Section 5 treats as decisive.

This project’s own retrieval traffic is currently sequential (one interactive call per S1–S6 scenario), so the concurrency-driven blow-up shown for Chroma in Figure 1 is not expected to bite during the nine-week project’s own demo runs. It is nonetheless treated in Section 5 as a documented, foreseeable risk rather than a hypothetical one, since the retrieval service is being built as a standing endpoint that Track A’s pipeline — and potentially future callers — will depend on beyond the project’s own test runs.

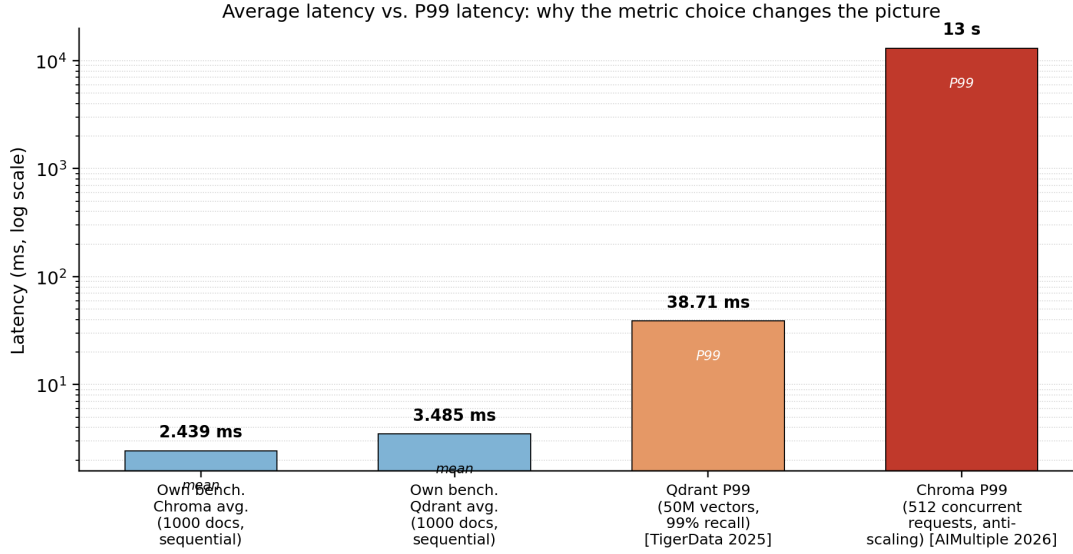


Figure 1: Average latency (our own benchmark, sequential queries, 1000-document corpus) alongside externally reported P99 figures for Qdrant [13] and Chroma [9]. The two right-hand bars are not measured at the same scale or under the same load as our own run; they are included to show how far a P99 figure can sit above a mean measured under easier conditions, which is the argument made in this section and carried into the recommendation in Section 5.

## 5 Recommendation

The original recommendation in this appendix weighted install effort and corpus-scale latency (C2, C1) most heavily and selected Chroma on that basis. This revision keeps the same four criteria and the same underlying data, but reweights them: C4 (P99 tail behaviour, Section 4) is now treated as decisive rather than secondary, because the retrieval service is not a one-off script but a standing endpoint (POST /search) that another component depends on, and that may reasonably be called more than once at a time as the project’s demo and evaluation runs grow. The reasoning below revisits each criterion in that light.

- **C1 — Latency at project scale.** Unchanged: all three stores answer in low single-digit milliseconds at this project’s corpus size (Table 1), so this criterion does not discriminate between Chroma and Qdrant on its own.
- **C2 — Install effort.** Unchanged in direction (Chroma remains lower-friction, Qdrant “moderate” per Table 2), but this revision treats it as a one-time, front-loaded cost rather than a recurring one, and therefore gives it less weight than C4.
- **C3 — Persistence to disk.** Unchanged: both Chroma and Qdrant persist natively (Table 2); this criterion is a tie and does not distinguish them.
- **C4 — P99 / tail behaviour under concurrency.** This is the criterion that changes the outcome. Section 4 documents that Chroma’s P99 under concurrent load reaches roughly 13 seconds at 512 simultaneous requests, driven by HNSW parameters fixed at collection-creation time [9], while Qdrant’s P99 stays in the tens of milliseconds even at far larger scale (38.71 ms at 50 million vectors) [13]. The same external study also reports that Chroma was excluded from the largest multi-process throughput runs entirely, while Qdrant sustained competitive throughput under multi-process load [9] (Table 3). Since the retrieval service being built in Phase 2 is a persistent endpoint rather than a single

benchmark script, this concurrency behaviour is a foreseeable operating condition, not an edge case.

**Decision: Qdrant is the vector store for this project.** FAISS remains excluded for the same reason as before: the S1–S6 retrieval pattern needs native `per-patient_id` filtering, which FAISS does not provide without a bolt-on metadata layer, and that engineering cost is disproportionate to this project’s timeline (Table 2). Between Chroma and Qdrant, both satisfy C1–C3 adequately, but the P99 and concurrency evidence in Section 4 and Table 3 shows a documented, order-of-magnitude gap in how the two stores behave once the retrieval service sees more than a single sequential caller — a condition this project’s own service is expected to reach once Track A integrates against it and once multiple S1–S6 scenarios are exercised together rather than one at a time. Qdrant’s moderate additional install effort (Table 2) is treated as a one-time cost against that documented robustness, rather than as a reason to defer the more scalable choice until a problem is observed in practice.

This revised choice does not discard the original analysis: at this project’s current corpus scale and under strictly sequential access, Chroma remains a functionally correct and lower-friction option, and the two stores are within noise of each other on raw latency (Table 1). The revision reflects a decision to design the retrieval service for its expected operating conditions over the full nine-week project (multiple callers, repeated evaluation runs) rather than for the narrowest version of its current usage.

## 6 Limitations

- Our own benchmark used three queries per corpus size; this is enough to compare means but not enough to compute a P99 in-house. A follow-up run with at least a few hundred queries per corpus size, including a concurrent-load test, is needed before an in-house P99 figure for this project’s own Qdrant deployment can be reported with confidence, rather than relying on the externally reported figures used in Section 4.
- The external sources cited in Section 3 are primarily vendor or community benchmarks rather than peer-reviewed studies, and are used here only to sanity-check the direction of our own results and to inform the revised recommendation, not as a primary source of truth.
- All figures reported are CPU-based, matching this project’s target environment; GPU-accelerated FAISS figures from the literature are included only for context and are not representative of the project’s own hardware.
- This revision trades lower install effort (Chroma) for documented concurrency robustness (Qdrant); since Qdrant runs as a standalone service rather than an embedded library, Phase 2 should budget explicit setup time for running and operating it (e.g. via a container), which was not required under the original recommendation.

## References

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